Newton's three laws

1st LAW

Inertia



If no net force acts, an object **keeps doing what it was doing**: at rest, or moving in a straight line at constant speed.

When a bus brakes, you keep moving forward.

2nd LAW

Force and acceleration



$$F = m \cdot a$$

The bigger the force, the bigger the acceleration. The more mass, the harder it is to accelerate.

A full trolley is harder to push than an empty one.

3rd LAW

Action and reaction



If A pushes B, B pushes A with a force of the **same size** in the **opposite direction**. They act on different bodies.

Rockets push gas down; the gas pushes the rocket up 🚀